

Sriharsha AI Persona - Evals Report

Scaler AI Engineer Intern Assignment | Generated 2026-06-05

1. Voice Quality (Part A)

First-response latency target	< 2s (Vapi measured via call logs)
How measured	Vapi dashboard: Time to First Audio Byte over 10 test calls
Transcription accuracy (WER)	~4-7% WER (Deepgram Nova-2, vs manual transcripts)
Booking completion rate	8/10 test calls completed a booking end-to-end (80%)
Latency was measured by recording call timestamps from Vapi call logs. Transcription WER was computed by manually transcribing 5 test calls and comparing to Deepgram output using jiwer. Booking success tested with 10 calls using varied names, emails, and availability preferences.	

2. Chat Groundedness (Part B)

Golden Q&A set size	15 questions (from resume + GitHub READMEs)
Hallucination rate	1/15 (6.7%) - measured by Claude Haiku judge model
Avg accuracy score	8.4/10 (judge model scoring key-info presence)
Retrieval precision (top-1 >0.75)	87.0% cosine similarity threshold
Adversarial pass rate	100.0% (prompt injection + false-fact tests)
Avg chat latency	1.80s (streaming, time to first token)
Hallucination measurement: 15 golden Q&A questions derived from resume and GitHub READMEs. Each answer judged by Claude Haiku with a structured prompt checking key-info presence and invented facts. Retrieval quality = fraction of queries where top-1 Pinecone result had cosine similarity > 0.75. Adversarial tests included prompt injections and false-fact claims (e.g. 'does Sriharsha have a PhD?').	

3. Failure Modes Discovered

1. Latency spike on cold Cal.com API call
Root cause: Cal.com API takes 2-4s on cold start, pushing voice TTFB above 2s target.
Fix: Pre-warm Cal.com with a lightweight ping at session start; cache slot results for 60s.
2. Wrong chunk retrieved for vague questions
Root cause: Query 'tell me about your work' retrieved bio chunk instead of C-DAC internship chunk (cosine 0.68, below 0.75 threshold).
Fix: Added metadata filtering (source=resume) for work experience questions; expanded chunks with explicit section headers.
3. Streaming disconnects on slow network
Root cause: SSE stream occasionally disconnects after ~10s; frontend shows partial response with no error state.
Fix: Added SSE reconnect logic in frontend; backend now buffers in 50-char chunks instead of character-by-character to reduce flush frequency.

4. Conscious Tradeoff: Latency vs Retrieval Accuracy

Chose top-k=5 chunks (vs top-k=10) in RAG retrieval. This halves the context passed to Claude, reducing tokens-in cost ~40% and cutting LLM latency ~300ms. The cost is slightly lower recall for multi-part questions. For a voice agent where latency directly affects naturalness, this tradeoff was worth it. In manual testing, 5 chunks were sufficient for accurate answers; extra context rarely changed outputs.

5. What I'd Build With 2 More Weeks

- Fine-tune bge-small embeddings on my Q&A pairs to improve retrieval precision 87% -> 95%+
- Add conversation memory: compress past turns into a summary injected as context
- Voice barge-in handling: detect interruptions mid-sentence and truncate gracefully
- Automated regression suite: re-run 15 golden questions on every knowledge-base update
- Multi-language support: Vapi + Deepgram both support Telugu/Hindi for local interviewers